



INTRODUCTION TO GAMES

Part 8: Playtesting & Evaluation

Today's Lecture: Playtesting & Evaluation

1. What is playtesting?
2. Why playtesting?
3. Playtesting methods.
4. Playtests in practice.
5. Case study: games research using gaze.

1.

What is Playtesting?



A definition.



PLAYTESTING

A comparison of **what it is** and **what it is not**.

Playtesting Goals

What it is

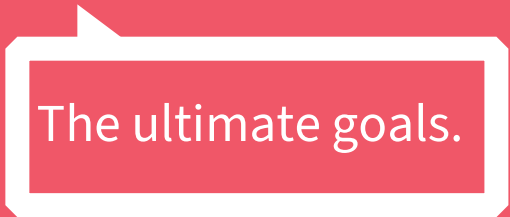
- An essential part of the **design process** (game design, level design, graphic design, etc.)
- A tool to evaluate the **game experience** (GX), focusing on the design.

What it is not

- It is **not primarily quality assurance** for bugs (although, in the industry, it's sometimes just that).
- It is **not a marketing survey tool** for potential target groups.

2.

Why Playtesting?



The ultimate goals.

Why Playtesting?



Identify Game Design Issues

Are the **core mechanics** of the game working? Is the experience designed appropriately?



Identify Level Design Issues

Do players **understand** the level? Do they reach their goals? Where do they fail?



Evaluate Aroused Emotions

How do the players **feel** while playing? Are the evoked emotions consistent with the game design?



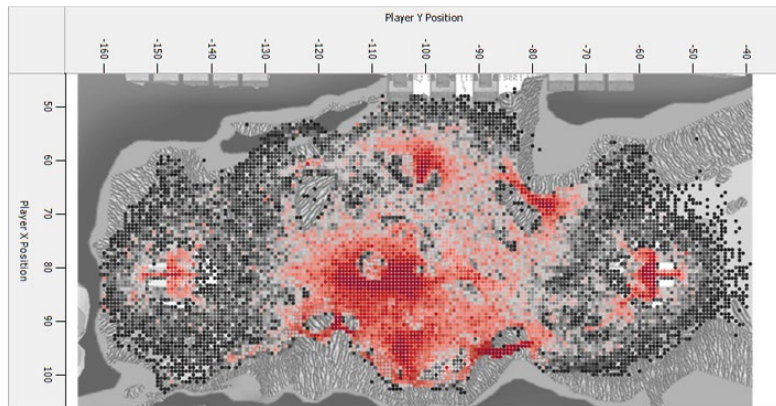
Evaluate Interface & Controls

Does the interface (visual and controls) fulfill its purpose? Do players get the appropriate **feedback**?

Halo 3 Playtests

Microsoft Games Research conducted over **3,000 hours** of playtesting with **600 players**.

This requires a big **budget**.



Indie Game Playtest

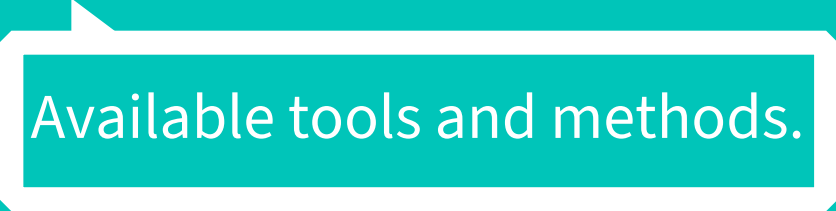
What should we do if it's just a tiny game? Maybe we should playtest it with 20 people in an office.

Playtesting **scales** well.



3.

Playtesting Methods

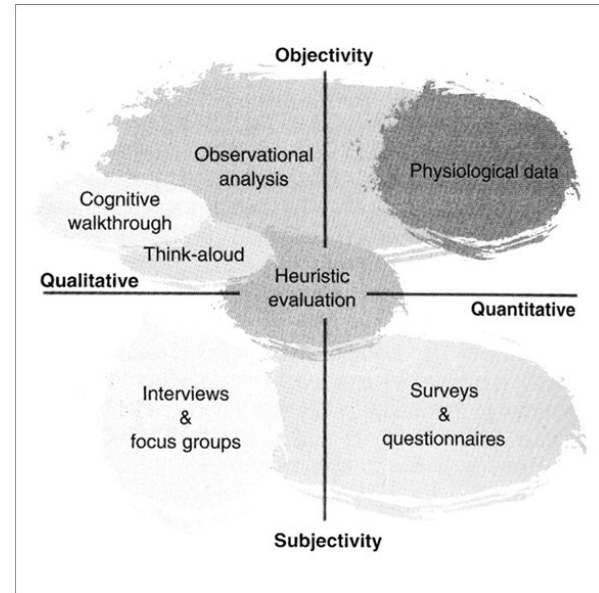


Available tools and methods.

Method Continuum

Current methods for evaluating entertainment technologies.

Methods differ between **objective** and **subjective**, and **qualitative** and **quantitative**.





METHODS

Four methods in detail: questionnaires, focus groups, physiological measurements, and heuristic evaluation.

Questionnaires

Gather information

Attitudes, values, behavior, etc.

- Quantitative questionnaires
- Qualitative questionnaires

Game Experience Questionnaire

Modular questionnaire for inquiring about the game experience.

Consists of multiple modules:

- Core Module
- Social Presence Module
- Post-game Module

Game Experience Questionnaire

Core Module: 33 items

7 components

- Immersion
- Flow
- Competence
- Positive & negative affect
- Tension
- Challenge

	not at all	slightly	moderately	fairly	extremely
I felt competent					

Questionnaires: Pro & Cons

Benefits

- Efficient tool.
- High number of subjects.
- Statistical analysis.
- No technical equipment is required.

Drawbacks

- Questionnaires and interactive products?
- After playing (not during).
- Development costs many resources.
- ... and requires knowledge.

Focus Groups

Powell et al. (1996)

- Discussion: a group of individuals
- Discussion: specific topic (comments, experience)
- Group size: 6 to 10 people
- Duration: 1 to 2 hours
- Social interaction leads to results
- Researcher: moderator

Focus Groups

An **explorative** method where little is known about the **topic**. The goal is to generate **hypotheses**.

Useful at the beginning to **test ideas** or to **evaluate** features.



Focus Groups: Pro & Cons

Benefits

- Key feature: social interaction.
- Words and actions are related.
- Defend and discuss ideas.
- Identify the importance of a topic.
- Subjects: a feeling of having accomplished something.

Drawbacks

- Social interaction.
- Researcher: little control.
- Opinion leader.
- Small group (general implications?).
- Subjects not anonymous.

Bioshock Focus Groups

Ken Levine (BioShock creator):

The focus test guy sort of patted me on the back and said, “Sorry, this game is going to be a failure.”



Physiological Measurements

Origins: biofeedback therapy & lie detectors

- Physiological parameters are measured.
- Evaluate specific game situations.



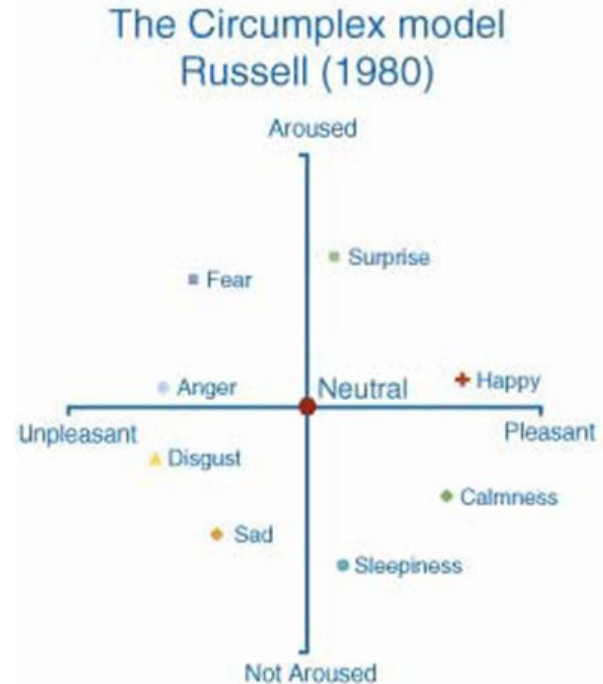
Physiological Measurements

- **Electrodermal Activity (EDA; old: Galvanic Skin Response, GSR)**
Skin Conductance
- **Electromyography (EMG)**
Muscle activity
- **Electrocardiogram (ECG)**
Heart rate (or more)
- **Electroencephalography (EEG)**
Brain activity

Physiological Measurements

Mapping parameters

- Emotion theory model
- Each emotion located on two dimensions
- Arousal (intensity)
- Valence (quality)

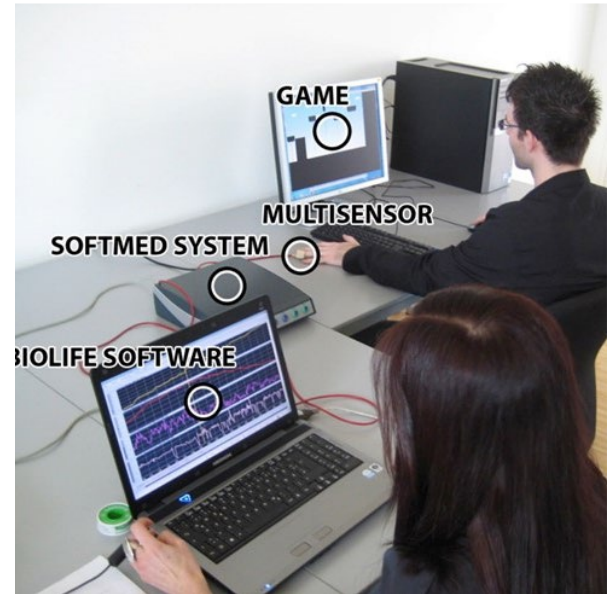


A Super Mario Bros.-style platformer scene. A small character with red hair and a purple dress is jumping over a gap. To the left is a large black block with a white diamond pattern. To the right is a large black block with a white diamond pattern. A speech bubble with a white border contains the text "Valory Physiological Measurements". The background features rolling green hills, small trees, and a yellow sky.

Valory
Physiological
Measurements

Valory: Procedure

1. Tutorial (how to interact with the game)
2. Listen to meditative music (3 min) to get low arousal
3. Sensors attached – baseline phase
4. Prototype A or B played; data recorded (3 mins)
5. Sensors detached, meditative music
6. Second prototype
7. Questionnaire

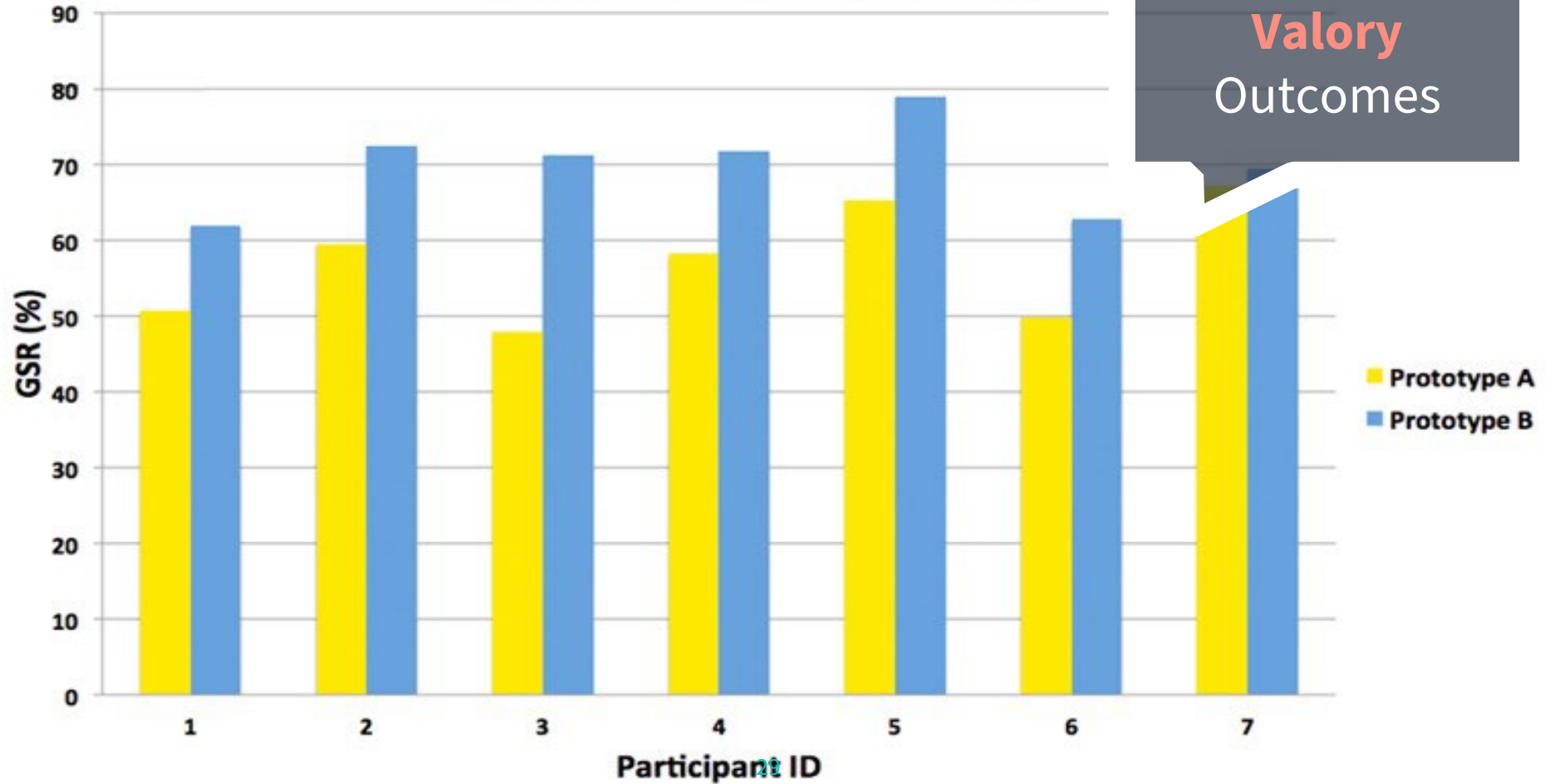


Valory: Outcome

- GSR data was gathered on a fixed interval basis.
- Sample rate: 1 sample per second.
- Mean values: calculated per participant and prototype scenario.
- GSR values: share a high individual variability.
- Measured values in prototype A: 1.01 μ S (minimum value)...
- ...to 12.98 μ S (maximum value)

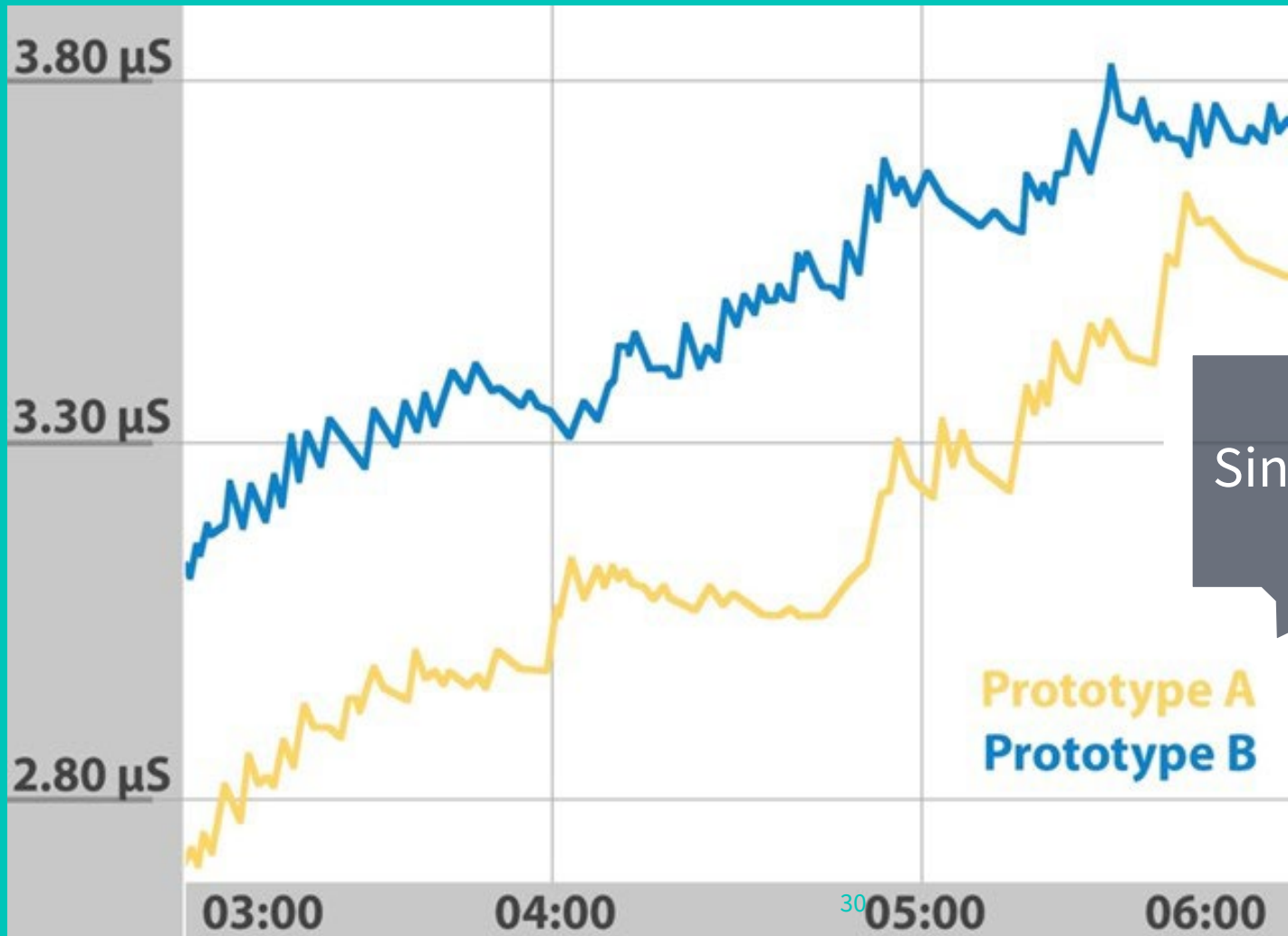
$$GSR_{normalized} = \frac{(GSR(t) - GSR_{min})}{(GSR_{max} - GSR_{min})} \cdot 100$$

Galvanic Skin Response (normalized)



Valory

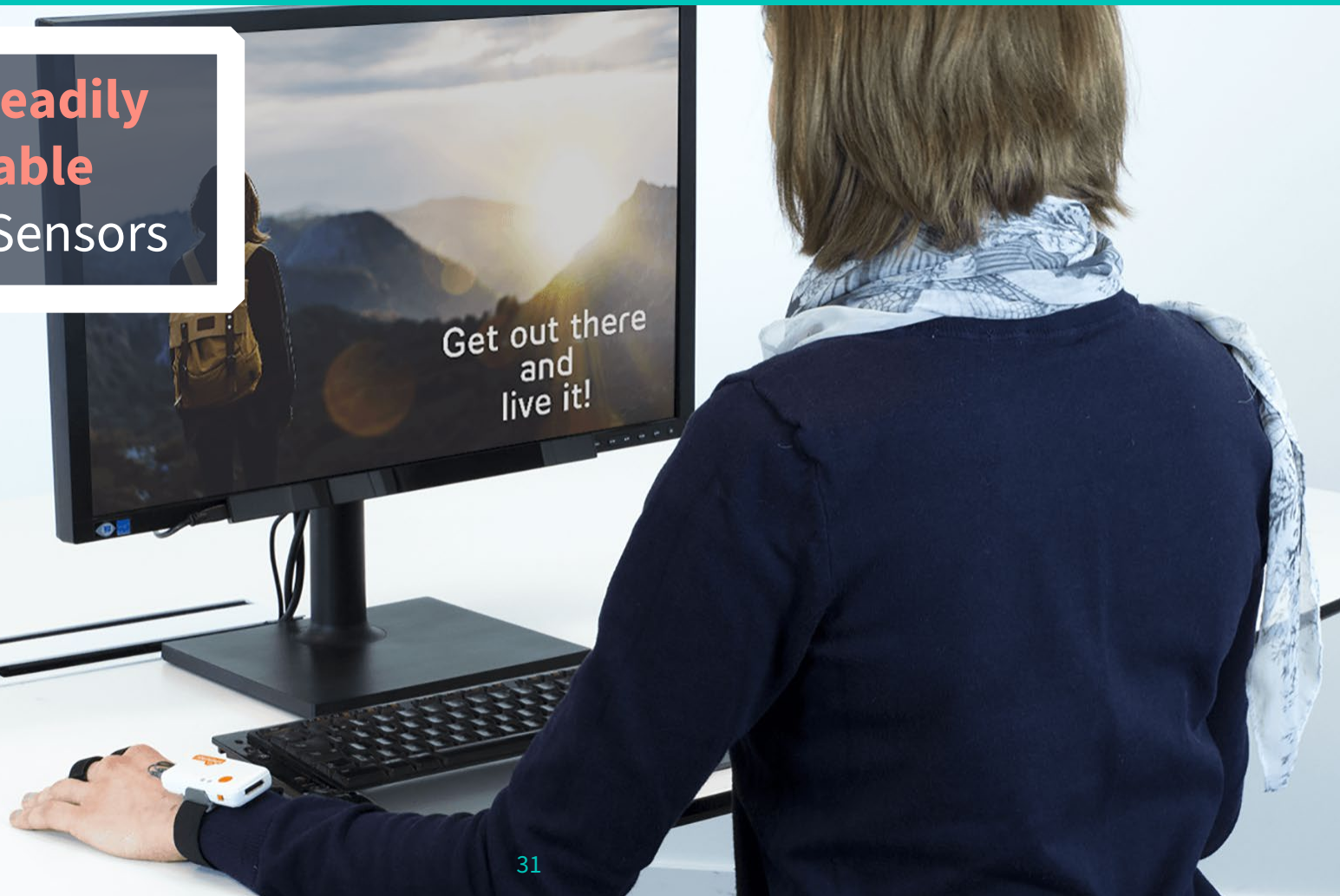
Outcomes



Valory
Single Participant
Diagram

Prototype A
Prototype B

**GSR is readily
available**
Smaller Sensors



Physiological Measurements: Pro & Cons

Benefits

- Evaluate during game sessions.
- Objective data.
- Reduction of artifacts (interviewer, questionnaire, etc.).

Drawbacks

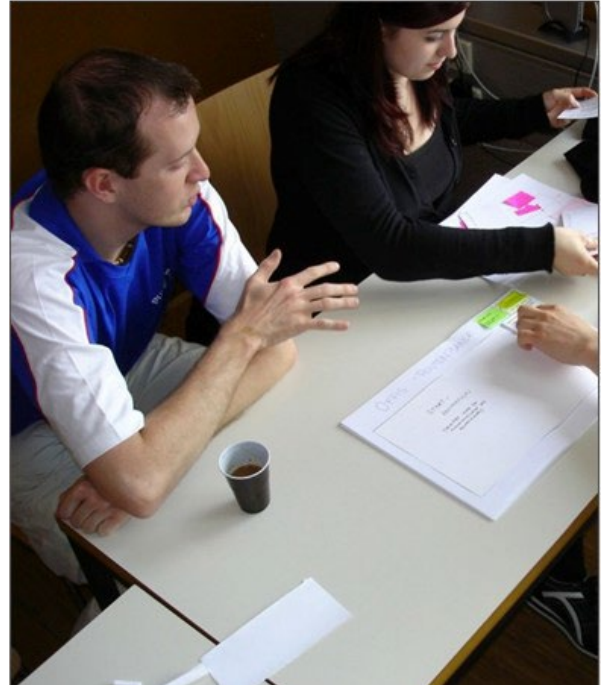
- Interpretation of data.
- Expensive technical equipment required.
- Sensors may disturb players.

Heuristic Evaluation

Heuristics are generalizations of design problems: **rules of thumb** for how things (the game, a UI, etc.) should be.

HE is an **expert evaluation method**, a tool for identifying problems.

No design standards!



Why Use Heuristic Evaluation?

Other methods are expensive and time-consuming.

E.g., full user studies and focus groups.

There is a need for a discount method.

It can be quick & dirty (if it works).

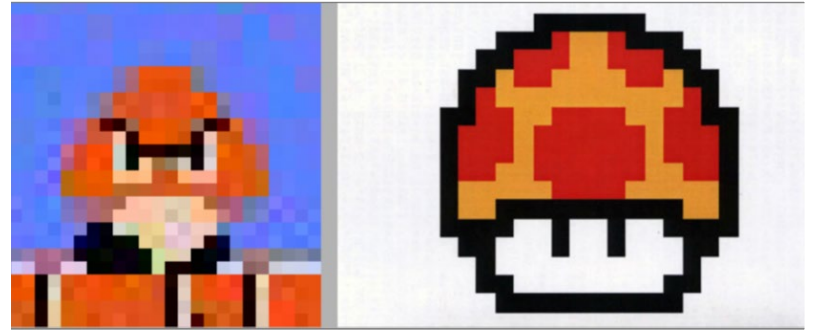
HE has a high practical value.

3 to 5 evaluators find up to 75% of all issues.

Examples for Heuristics

Players should be able to identify relevant game elements effectively (e.g., power ups).

Goomba vs. Mushroom



Examples for Heuristics

The level's goal has to be clear to players.

Half-Life 2: Ravenholm



Heuristic Evaluation: Procedure

1. The game is evaluated by 3 to 5 experts.
2. Experts use a list of heuristics.
3. Experts check if the heuristics are violated.
4. The outcome is a list of violated heuristics.
5. The list contains the type of problem and a severity rating.

0 = I don't agree that this is a usability problem at all.

1 = Cosmetic problem only: need not be fixed unless extra time is available on project.

2 = Minor usability problem: fixing this should be given low priority.

3 = Major usability problem: important to fix, so should be given high priority.

4 = Usability catastrophe: imperative to fix this before product can be released.

Heuristic Evaluation: Procedure

1. Expert lists are combined to create a **consolidated file**.
2. Only problems identified by **at least two experts** are selected.
3. The final step is a **report**.
4. Problems are **sorted by severity**.
5. The report should include **comments** rather than solutions.

Heuristics in Practice: PLAY

Game Usability Heuristics (PLAY) for Evaluating and Designing Better Games: The Next Iteration

https://link.springer.com/content/pdf/10.1007/978-3-642-02774-1_60.pdf

Game Usability Heuristics (PLAY) For Evaluating and Designing Better Games: The Next Iteration

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Abstract. Game developers have begun applying formal human-computer interaction (HCI) principles in design. Desurvire et al [2] adapted a set of Heuristics for productivity software to games. The resulting set, presented at CHI 2004, was *Heuristics to Evaluate Playability* (HEP). Generalization of these heuristics is required to make them applicable to a multiple of game genres and game deliveries. This follow-up study focused on the refined list, Heuristics of Playability (PLAY), that can be applied earlier in game development as well as aiding developers between formal usability/playability research during the development cycle. Heuristics were formed based on their efficacious scores on the popular game review website, metacritic.com. Fifty-four gamers rated High and Low ranked games on 116 potential heuristics. Implications for how these Heuristics will help developers improve game quality are discussed. PLAY has been found useful in design evaluation and elf-report survey format.

Heuristics in Practice: Shadowgrounds

An **expert evaluation** was applied to develop Frozenbyte's game **Shadowgrounds** (2006).

<https://web.archive.org/web/20220519234032/https://www.shadowgroundsgame.com/>

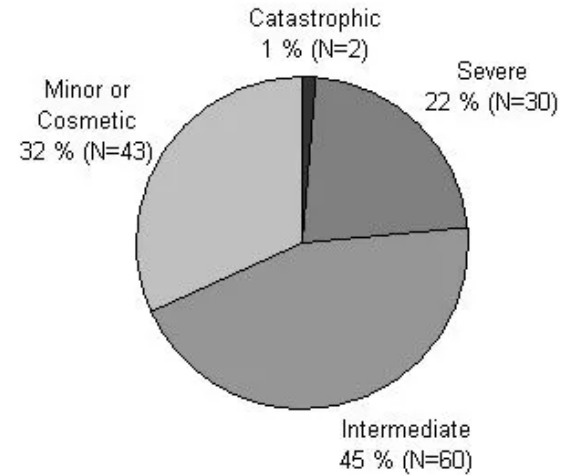


Heuristics in Practice: Shadowgrounds

Three experts reviewed the game for about a week.

135 problems were identified.

Problems found in the usability expert evaluation



Heuristics in Practice: Shadowgrounds

Problem

One color has multiple meanings in the map display.

Rating

Catastrophic

Description

The light blue color in the map display has three different meanings. It marks the area visited, unknown area, and areas that cannot be accessed (e.g., mountains).

Using the same color to symbolize three different things makes understanding the map difficult. Every time the user opens the map, s/he must stop for a second and think what the colors mean in different locations. There is also a danger that the users do not understand what the colors mean or misinterpret the map.



Solution

Use different colors for displaying visited, unknown and inaccessible areas.

Developers' comment

This problem could have been addressed earlier if the design document had covered the map screen more thoroughly. In the final game, a map legend will be displayed, and only one color (shade) will be used for one meaning.

Heuristics in Practice: Shadowgrounds

Problem

No feedback is given if the player cannot pick an item.

Rating

Severe

Description

Sometimes it happens that the player cannot pick up an item because there is no room in the inventory. If this happens, the user is not given any feedback.

This is problematic as the user may not know why s/he cannot pick up the item. It is likely that the user will figure it out eventually, but the confusion and extra effort required are likely to cause frustration.



Solution

Give the user proper feedback in every situation where the user interacts with the environment. If the item cannot be picked up, inform the user about this with a sound and/or textual feedback.

Developers' comment

This was addressed only vaguely in the design document and was overlooked by the programmers at the time. It is likely that this feature would have been implemented in due time and goes to show that sometimes developers leave many of the smaller usability issues to the final stretch of development.

Heuristics in Practice: Shadowgrounds

Problem

It is not immediately obvious whether the upgrades presented next to the weapon are installed or not.

Rating

Intermediate

Description

The upgrades available for the weapons look active even before purchasing them. This is because they are marked with a similar color scheme as the weapons next to them, which are active.

This is especially problematic early in the game. Because the upgrades look active, the user may not understand that this menu can be used to purchase new upgrades.



Solution

Make the upgrades that have not been purchased yet clearly inactive. This can be done for example by using a darker shade of blue to present the upgrades that have not been bought yet.

Developers' comment

The weapon upgrade feature had not been fully implemented by the time of the testing, but regardless, this particular problem had not been taken into consideration. Later user-testing showed that this was definitely a problem with gamers, especially in the beginning of the game.

Heuristics in Practice: Shadowgrounds

Problem

Red lights in the computers are easily mixed with enemies displayed in the radar.

Rating

Minor

Description

Enemies are marked with small red spots in the radar. The radar is semi-transparent: the scenery is visible through it.

In some buildings there are computers and other equipment that have small red lights. When it happens that such a light is visible through the radar it is easily thought to be an enemy.



Solution

Do not use small red lights in levels, or change the method of displaying enemies in the radar.

Developers' comment

Sometimes the developer can freely ignore the comments or suggestions made by the usability report. Here, Frozenbyte decided that this problem occurs too rarely to warrant any changes to the user interface.

Heuristic Evaluation: Pro & Cons

Benefits

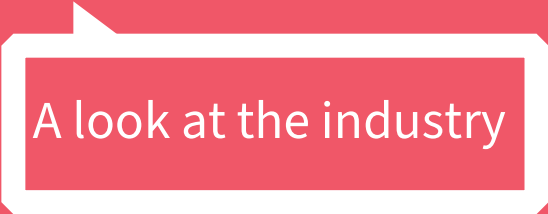
- Easy & cheap.
- What you need: 3 to 5 domain experts, a good set of heuristics, and your game.
- Can be used early in the design process.

Drawbacks

- No scientific value (low number of subjects, etc.).
- Typically, not all problems can be found.
- Dependent on the quality of heuristics.

4.

Playtests in Practice



A look at the industry

Playtests in Practice

Playtesting is usually a mix of methods.

- Observation
- Interviews
- Physiological measurements



Playtests: Pro & Cons

Benefits

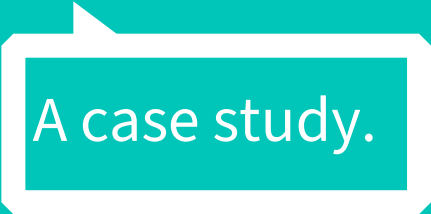
- A detailed picture of the interaction process.
- Players can give immediate feedback.
- Various channels of information.
- Lots of data.
- Address different problem areas.

Drawbacks

- (Really) expensive.
- Takes time.
- Requires special technical equipment.
- Lots of data.
- Can only simulate the typical usage.

5.

Games Research using Gaze



A case study.

Gaming & Gaze

Eyetrackers for **gaze-based interaction** are now readily available for use in games. Either for **input** or for **evaluation** purposes.



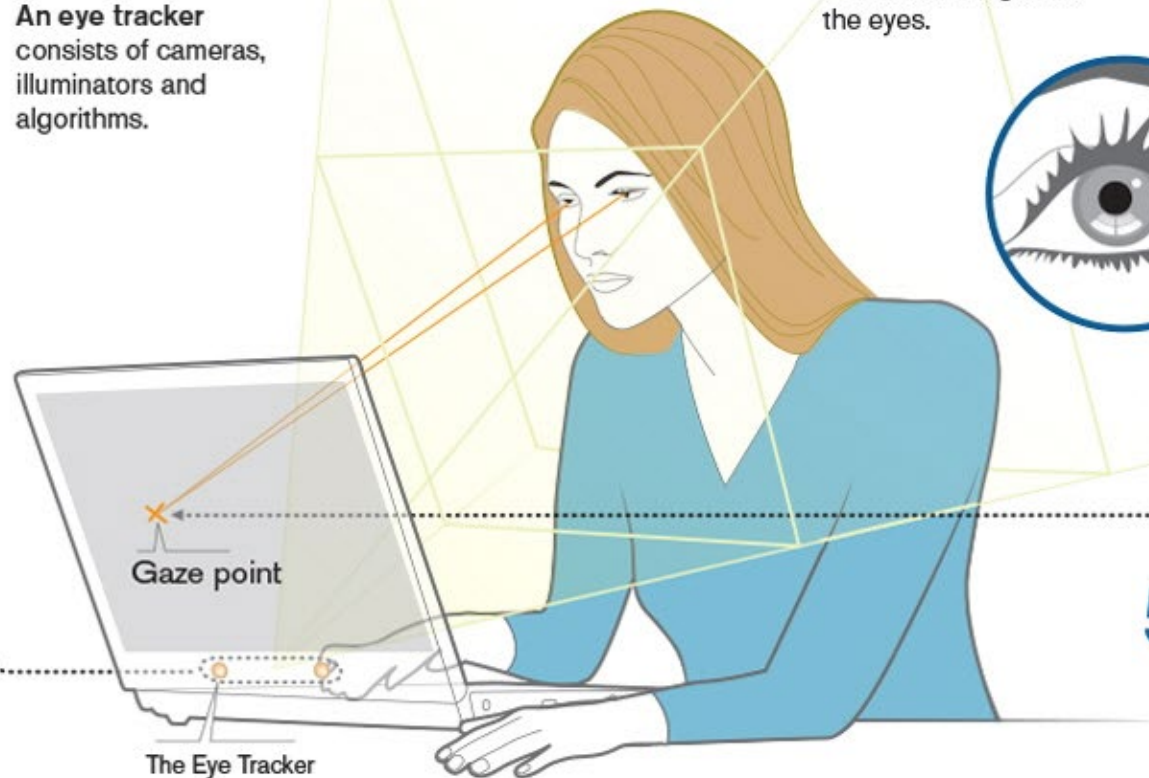
1 An eye tracker consists of cameras, illuminators and algorithms.

2 The illuminators create a pattern of near-infrared light on the eyes.

3 The cameras take high-resolution images of the user's eyes and the patterns.

4 The image processing algorithms find specific details in the user's eyes and reflections patterns.

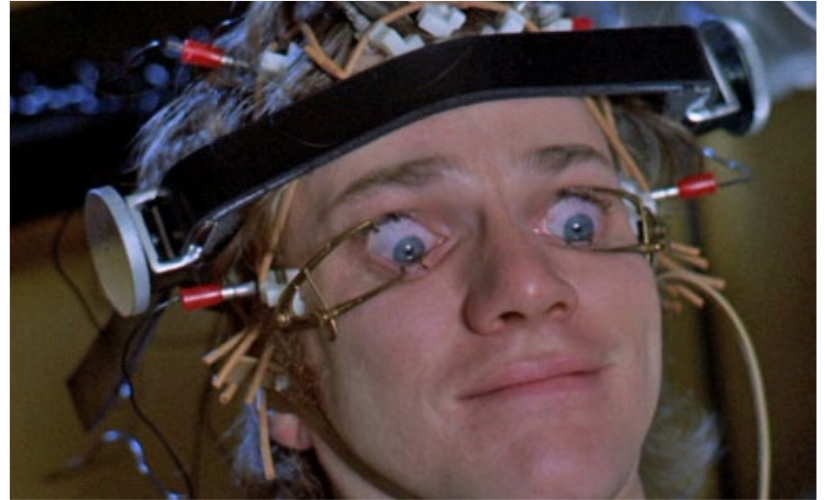
5 Based on these details the eyes' position and gaze point are calculated, for instance on a computer monitor, using a sophisticated 3D eye model algorithm.



What does eyetracking do?

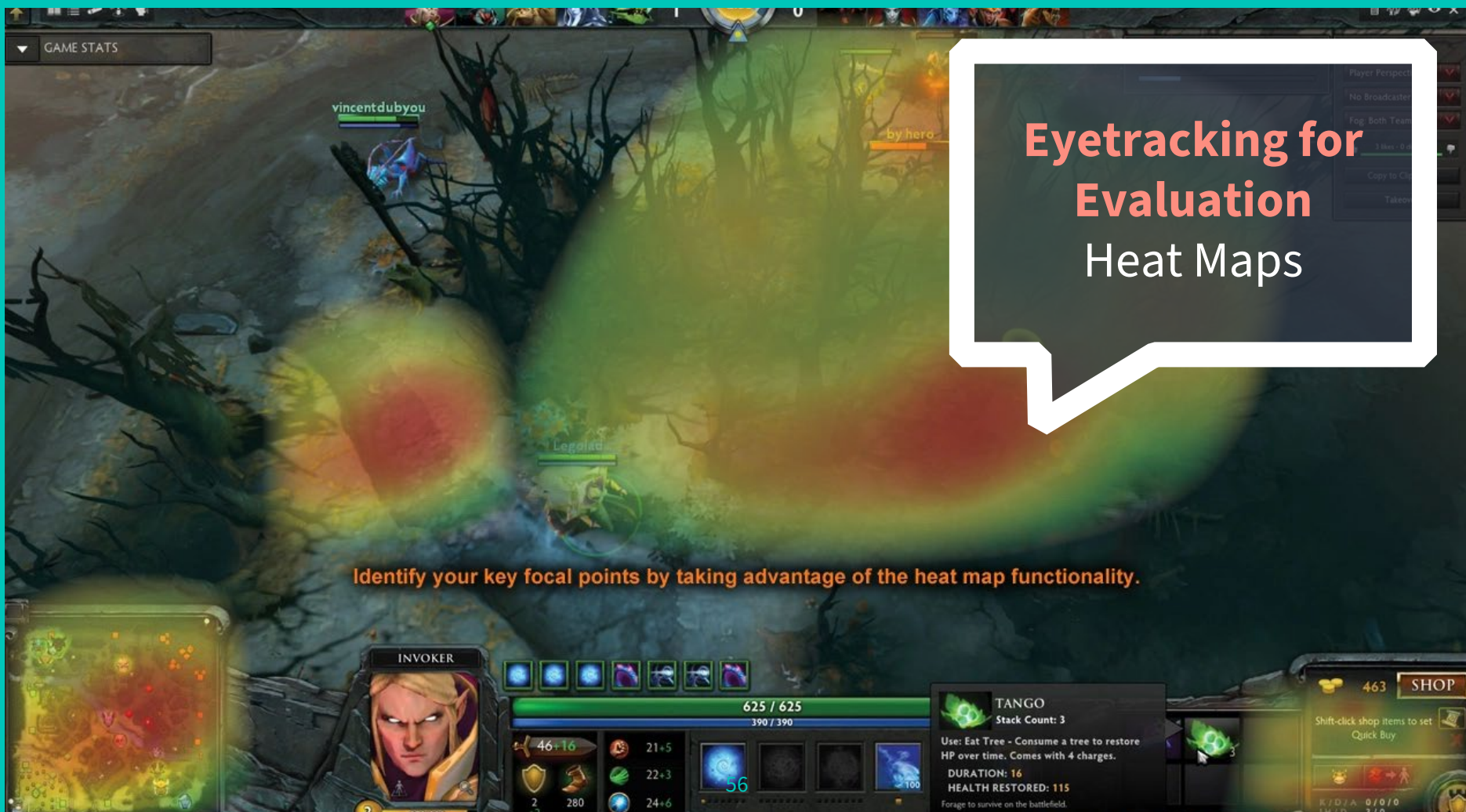
In general, eye-tracking is a process that **records eye movements**, allowing us to determine where an observer's gaze is fixed at a given time.

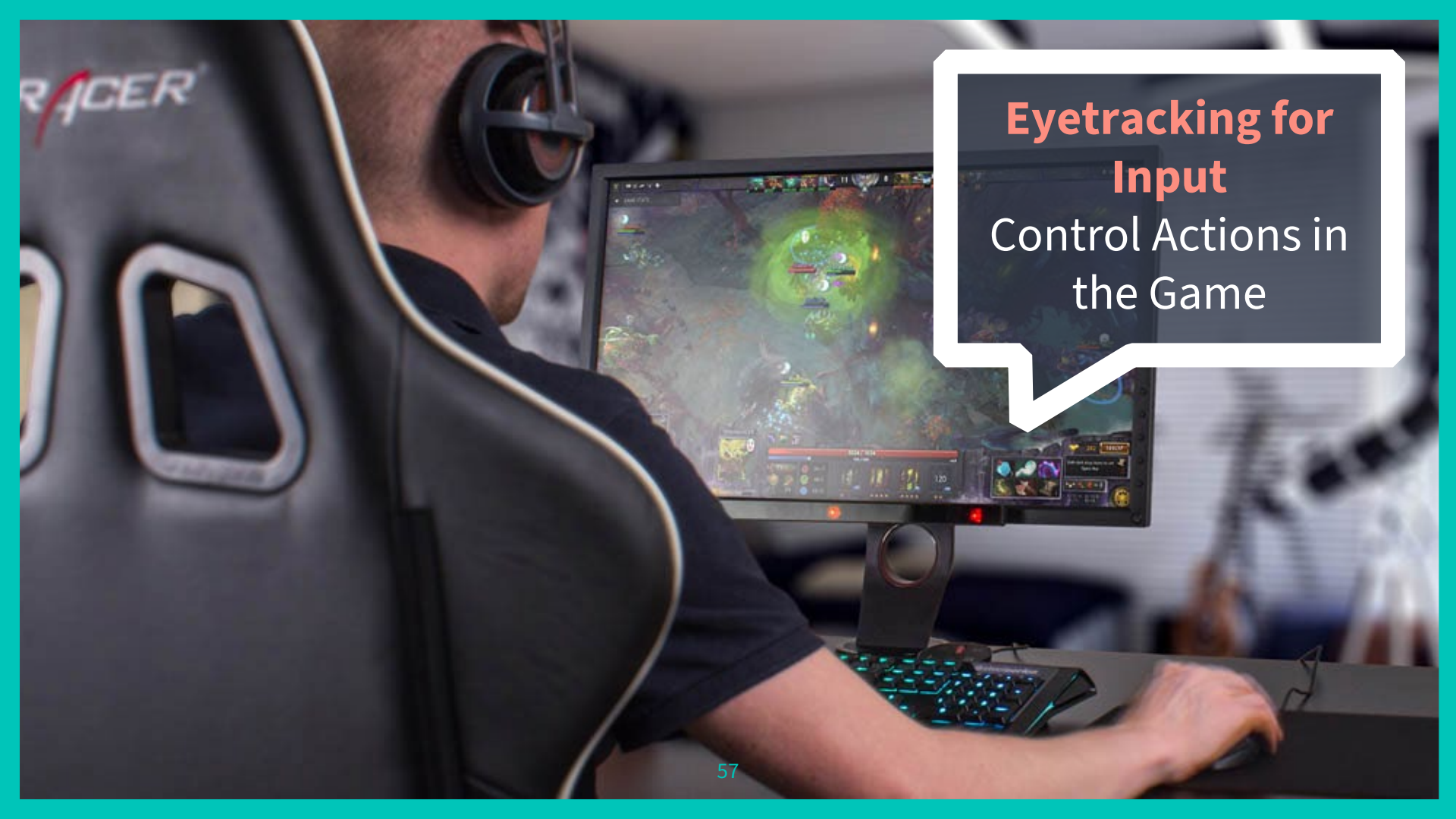
The point being focused on a screen is called a **gaze point**.



Eyetracking for Evaluation Heat Maps

Identify your key focal points by taking advantage of the heat map functionality.



A person is shown from the side, wearing a black headset and a black t-shirt, sitting in a black and white racing-style gaming chair. They are looking at a computer monitor displaying a game. The chair has the 'RACER' logo on the backrest. The monitor shows a game interface with a central green circular area, various icons, and a health bar. A speech bubble with a white border is overlaid on the right side of the image, containing text about eyetracking for input control actions in the game.

Eyetracking for Input Control Actions in the Game

Lots of Gadgets
Tobii Eye Tracker 5



Lots of Gadgets
Pimax Crystal Super
(VR + Eyetracking)

State of the Art
Foveated Rendering

the focus area can be reduced

Footage: Laubox



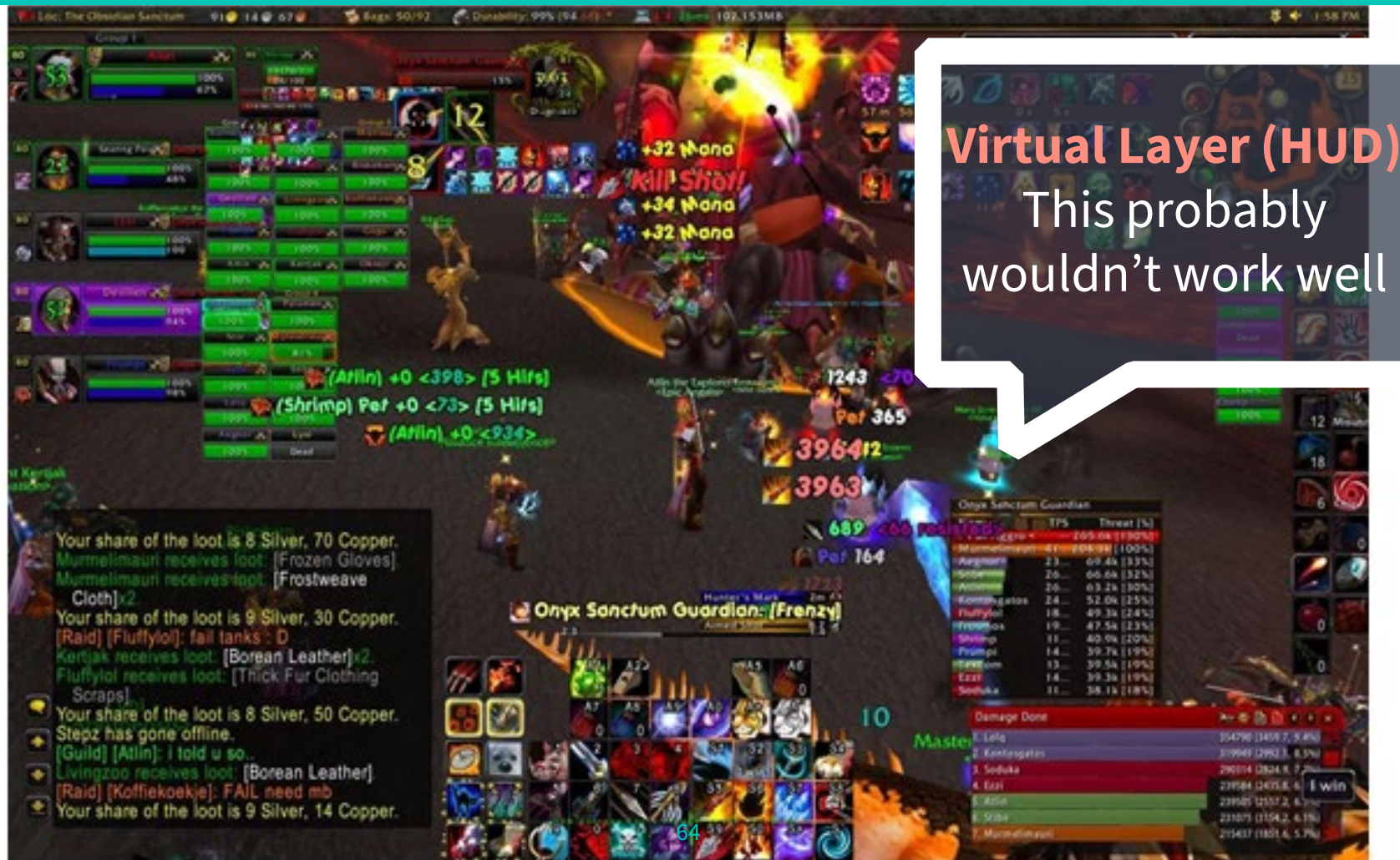
PROBLEMS?

When using gaze input, specific **issues** must be addressed.



Interface and Gaze

Selecting things



Virtual Layer (HUD)

This is probably better

AUX POWER
■■■■■■■■■■

HEALTH 97

SUIT 33

65

AMMO 16

111



Diegetic Interface

Looking at an
interface vs.
interacting with it

How to Design Interfaces for Gaze?

There are usually two perspectives.

How to deal with interfaces?

Minimize them

Many interface designers aim to **minimize** the presence of UI components by integrating them into the game world.

Keep them

On the other hand, some researchers claim that such features **do not harm**, as they are necessary and have become a conventional form of communication.

Lots of Questions

Adding gaze to the issue:

- Should designers **strive for virtual interface elements** to make playful gaze interaction more pleasant and practical?
- Or should the interface design **conceal information** regarding the gaze interaction (position, dwell time, etc.)?
- Does the **game genre** play a relevant role in this consideration?

Let's answer it using the concept of **presence**!



“

Does the gaze-related virtual interface information (e.g., the position of the player's gaze via visual elements) have an impact on the player's presence?

Presence in Games Research

- **Hartescu and Oikonomou (2011)**
Gaze input fosters presence
- **Schultze, 2010**
Sense of existing or of “being there” in a virtual space
- **Tamborini & Skalski, 2010**
Composed of the three dimensions

Social presence, spatial presence, and self-presence



ANSWERS!

Presence in games was tested through two game prototypes.



FRACTILE

Fractile

A game prototype
about memories
and war crimes

Limus and the Eyes of the Beholder

**Limus and the Eyes
of the Beholders**
Super Mario World
with Gaze



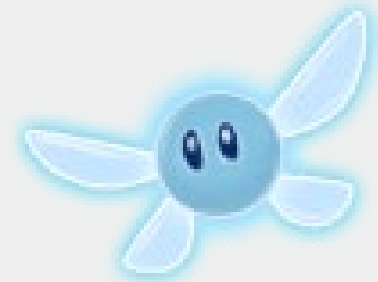
The Actors



Limus



Spikees



Lumees

Fractile: The Two Testing Scenarios

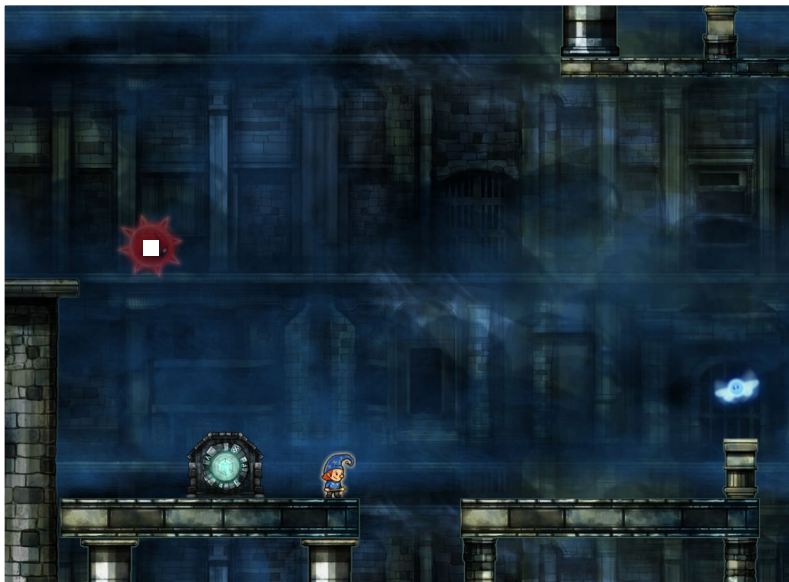


With Rectangle

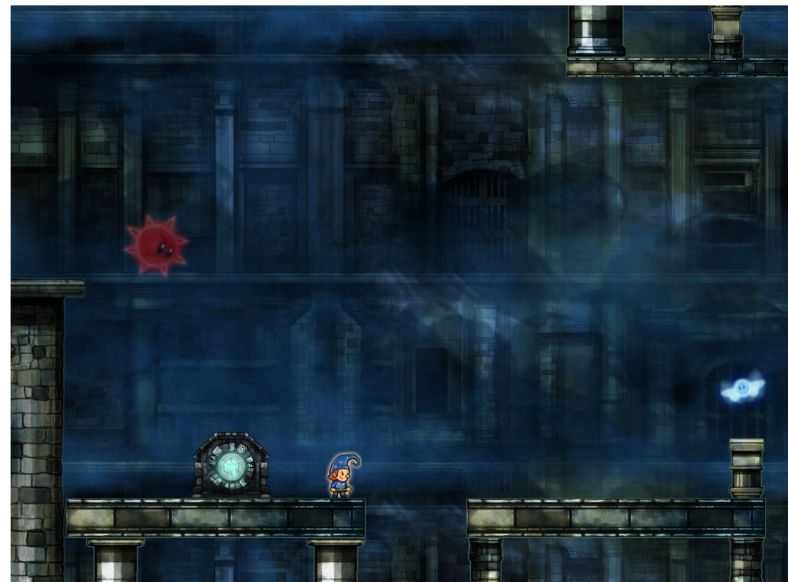


Without Rectangle

Limus: The Two Testing Scenarios



With Rectangle



Without Rectangle

SPSL: Spatial Presence: Self Location

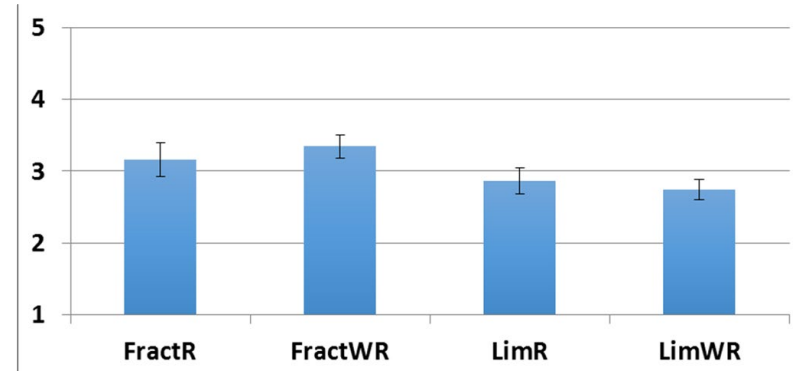
- Part of the MEC-SPQ (Measurement, Effects, Conditions Spatial Presence Questionnaire) by Vorderer et al. (2004)
- Reflects the physical quality of presence
- Example: “I had the feeling that I was in the middle of the action rather than merely observing.”

SPPA: Spatial Presence Possible Actions

- Also, part of MEC-SPQ by Vorderer et al. (2004)
- Assesses the impression of being able to act
- Example: “The objects in the presentation gave me the feeling that I could do things with them.”

Study and Results of the SPSL

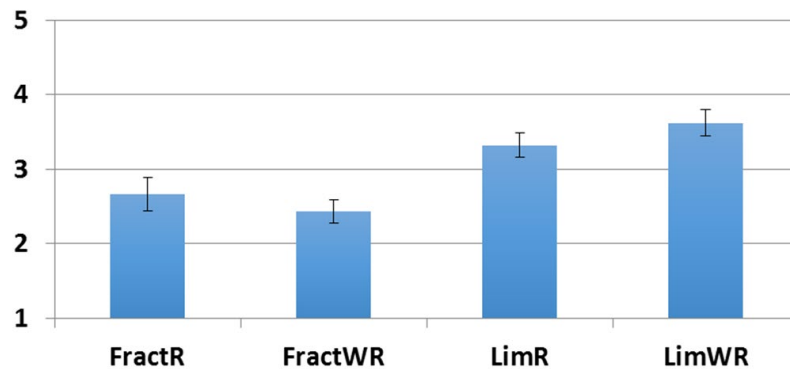
A study with 24 participants showed that the version **without the rectangle** hat had the highest presence but differed in the genres (Limus/Fractile).



Study and Results of the SPPA

Concerning possible actions, Limus had far better ratings than the 3D prototype Fractile.

Genre does have a significant impact!



More findings

Perspective and genre strongly influenced presence.

Players were annoyed by the rectangle but got used to it.

In Limus, it gave them the feeling of being a godlike creature that must protect the wizard.

Credits

- S. 8: Eye tracking games – Killzone 3: <https://www.youtube.com/watch?v=YwGMqVvJKxU>
- S. 9: Halo 3 Playtests: <http://halo.bungie.org/bwu/index.html?item=128>
- S. 10: Fray Playtests: <https://www.moddb.com/games/fray/news/fray-war-tears-and-playtests1>
- S. 21: Bioshock Focus Groups: <http://www.rollingstone.com/culture/news/we-were-all-miserable-inside-bioshock-video-game-franchise-w439921>
- S. 31: Shimmer3 GSR Sensor: <https://www.tobii.com/products/accessories/shimmer3-gsr-unit>
- S. 32: EMOTIV EPOC X: <https://www.emotiv.com/products/epoc-x>
- S. 38: Half-Life 2: Ravenholm: <https://valve.fandom.com/de/wiki/Ravenholm>
- S. 41: Shadowgrounds: <https://www.gog.com/de/game/shadowgrounds>
- S. 42: Shadowground HE: <https://www.gamedeveloper.com/production/better-games-through-usability-evaluation-and-testing>
- S. 49: Halo 3 Playtesting: http://gamemodworkshop.com/readings/Playtesting_Game_Design_Workshop.pdf
- S. 50: Bungie playtesting Halo 3: <https://www.youtube.com/watch?v=OZVp-9pUwyk>
- S. 53: Eyetracking for games: <https://hight3ch.com/worlds-first-eye-tracking-gaming-gear/>
- S. 54: Eyetracking calibration: <https://help.tobii.com/hc/en-us/articles/360023794433-How-does-the-calibration-work->
- S. 55: Clockwork Orange: <https://www.puremovies.co.uk/columns/the-case-of-a-clockwork-orange/>

Credits

- S. 56: Tobii Heatmap: <https://techcrunch.com/2014/06/09/steelseries-teams-with-tobii-to-track-your-eyes-and-make-you-a-better-gamer/>
- S. 57: Steel Series Eyetracker: https://www.legitreviews.com/future-streaming-unveiled-steelseries-sentry-eye-tracker_155942
- S. 58: Assassin's Creed Valhalla Eye Tracking: <https://www.youtube.com/watch?v=ID5GjiLhDDA>
- S. 59: Tobii Eye Tracker 5: <https://gaming.tobii.com/product/eye-tracker-5/>
- S. 60: Pimax Crystal Super (50 PPD): <https://www.aerosoft.com/de/shop/move/hardware/4551/pimax-crystal-super-50-ppd>
- S. 61: Foveated Rendering: <https://eu.pimax.com/blogs/highlights/why-dynamic-foveated-rendering-matters-for-vr>
- S. 63: Heavy Rain: <http://vignette4.wikia.nocookie.net/heavyrain/images/d/da/Aripic.png/revision/latest?cb=20100209175303>
- S. 64: World of Warcraft: https://www.reddit.com/r/wow/comments/14blxe/i_was_sifting_through_some_old_screenshots_from/
- S. 66: Dead Space 3: <http://images.eurogamer.net/2013/articles//a/1/5/5/4/2/6/2/eurogamer-amiybk.jpg>